

SRE, Cybersecurity and the Law of Technological Risk: Reliability, Security and Responsibility

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ABSTRACT

Oversight of digital infrastructures now demands a level of operational awareness that traditional legal methods were not really designed to handle. Systems no longer remain stable long enough to be examined from the outside. They evolve as they run, through automated deployments and data-driven adjustments that are often difficult to reconstruct after the fact.

Sometimes that reconstruction is simply not possible.

In learning environments, internal decision paths may shift without leaving a clear trace for occasional inspection. When supervision cannot follow at least part of this movement, it risks missing the moments in which routine disturbances begin to produce real harm.

This paper puts forward a practical approach to supervision that treats technical literacy as part of the ordinary responsibilities of public institutions. It builds on regulatory practices that have started to incorporate algorithmic audits and impact assessments into day-to-day oversight, as well as on traditions that accept the need to adapt methods to the systems being examined. The aim is not to replace technical expertise, but to make it usable in contexts where decisions still need to be justified in legal terms.

The focus remains deliberately grounded. Effective supervision depends on traceable data governance, inspection routines guided by operational risk and evidentiary standards for logs and metrics that can hold under scrutiny. It also depends on testing environments where failures can be reproduced without amplifying their impact.

And this is where cybersecurity becomes hard to separate from supervision.

The ability to detect, analyse and respond to incidents relies on the same conditions: integrity of data, availability of records and systems that can actually be observed while they run. None of this is useful unless supervisory teams are able to question engineers directly and understand the answers well enough to act on them, even if that understanding is only partial at first.

Taken together, these elements point to a form of institutional capacity that strengthens oversight without adding unnecessary layers of formalism. Reliability emerges not as a promise but as a way of working, one that accepts error as part of system behaviour and responds to it with discipline. The paper explores how such habits can take shape within legal institutions and how insights from Site Reliability Engineering can support a more grounded and accountable approach to technological risk.

KEYWORDS

algorithmic governance, cybersecurity governance, regulatory capacity, algorithmic audits, risk-based supervision, data governance and lineage, evidentiary standards for logs and metrics, incident response, red-team exercises, socio-technical risk

EXECUTIVE SUMMARY

The supervision of digital infrastructures has entered a phase in which traditional legal methods struggle to capture how modern systems actually operate. Automated pipelines, continuous deployments and data-driven adaptations mean that systems evolve while they

run, often in ways that are difficult to reconstruct afterwards. Learning models make this even harder by shifting internal decision paths in response to new inputs.

In practice, that movement is only partially visible.

Regulators who cannot follow at least part of it risk overlooking the moments in which routine disturbances begin to escalate into real harm. In this context, operational awareness and cybersecurity have become part of what can reasonably be expected from public responsibility.

Despite advances in algorithmic audits, impact assessments and risk-based supervision, most legal institutions still rely on policy declarations and post-hoc reviews. The ability to obtain technical evidence that can be tested and defended remains uneven.

And when that evidence is missing, the difference shows.

This gap makes it harder to distinguish ordinary fragility from negligence, particularly in security-related incidents. The paper proposes a capability model to help legal practitioners engage more directly with system behaviour, drawing on Site Reliability Engineering and its emphasis on monitoring, structured response and documented decision-making.

The connection with cybersecurity is not incidental.

The same evidentiary conditions underpin incident response, monitoring and adversarial testing.

The analysis draws on regulatory experiences in Europe, Singapore and Latin America and identifies key gaps in supervisory capacity, including limited fluency in system behaviour, uneven data governance practices and a persistent disconnect between legal and technical teams.

That disconnect is harder to resolve than it appears.

SRE practices provide a practical way of distinguishing failure from negligence. Where documentation is missing, institutional neglect becomes difficult to ignore.

Effective oversight depends less on new principles and more on institutions capable of learning over time. A system that cannot explain its failures is difficult to supervise. One that does not learn from them presents a different problem.

Reliability, in this sense, is not a promise but a way of working. The convergence of legal reasoning, engineering practice and cybersecurity offers a more realistic path for governing complex digital systems.

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A. INTRODUCTION

Discussions about technological reliability have reached a point where some of the older assumptions about risk in automated, high-stakes environments no longer hold up particularly well. Systems built on automation, orchestration and continuous delivery tend to treat error as part of their normal operation.

Not as an exception.

A failure is rarely a dramatic rupture. More often, it appears as a predictable disturbance that needs to be measured, contained and, ideally, used to improve the service that produced it. Legal analysis that still focuses on isolated faults struggles to capture how risk actually emerges in these architectures. It rarely does so in a single moment.

This tension has started to push engineers and legal practitioners closer together, sometimes out of necessity more than design. Across Latin America, concerns about algorithmic harm have encouraged supervisory bodies to adopt more proactive safeguards. In the United States, the NIST AI Risk Management Framework begins to outline what measurable expectations for automated systems might look like, even if its practical influence remains uneven and, in some contexts, still limited.

Following a system after it is deployed is not straightforward.

What these developments share is a growing awareness that duties of foresight, traceability and timely response cannot be satisfied through document review alone. Supervision has to reach the operational layer, where behaviour becomes visible rather than simply described (de Teffé and Medon, 2020).

Researchers have started to describe what is sometimes called a law of technological risk. It emerges from the practical reality that agency in modern systems is distributed across people, code and infrastructure. Causation becomes less a straight line and more an effort to reconstruct how design choices, operational routines and control mechanisms interacted over time.

In practice, reconstruction is rarely clean.

Developments in data protection and civil liability, particularly in Latin America, are already pushing legal doctrine in that direction. Governance duties, ex ante mitigation and clearer links between information, security and redress are becoming central

organising principles. Scholarship in personal data protection has emphasised that prevention and responsibility depend on objective organisational duties and user-centred safeguards, ideas that align quite naturally with technical practices such as observability and post-incident learning (Bioni and Dias, 2020).

The platform ecosystem makes these issues harder to ignore. Platforms curate content, automate ranking and deliver essential services at a scale where classical oversight tools start to feel insufficient. Work on intermediary liability in the region has argued for standards that reflect actual functions, exposure to risk and real capacities for control. Notice regimes, traceability and proportionate responses have become recurring themes.

Scale changes everything, including responsibility.

This body of doctrine gradually shifts the notion of reliability away from a purely technical aspiration and towards something closer to an institutional duty. It also changes how negligence should be understood when failures stem from structural weaknesses rather than from an individual's error (Palazzi, 2012).

The purpose of this section is to show how a culture of operational reliability can reshape the legal meaning of failure and negligence, and how it might support a verifiable standard of reasonable technical care. Properly understood, reliability is less a property of systems and more an organisational position. It depends on the ability to anticipate incidents, absorb them without multiplying harm and learn from them through documented processes that can be examined with some rigour.

This is harder to demonstrate than to claim.

Latin American administrative and civil law traditions already tie prevention and user protection to institutional responsibility, which provides a useful foundation for thinking about accountability in automated environments. The challenge is to accept the inevitability of error without weakening the duty of care, and to connect legal prevention with technical risk management in a way that treats error as evidence rather than as something to be concealed (de Teffé and Bodin de Moraes, 2017).

Courts and regulators will eventually have to confront a more uncomfortable point. Highly automated systems tend to blur the boundaries on which both civil liability and administrative oversight were built. A model of negligence that imagines a single actor

making a single mistake does not hold up well when architectures are distributed, deployments are continuous and decision pathways are often opaque.

Responsibility does not disappear, but it becomes harder to locate.

A law of technological risk will need to recalibrate its understanding of fault and diligence. In many cases, decisions about monitoring, documentation and recovery play a larger causal role than any individual technical slip. Administrative bodies already have tools that could be adapted to support this kind of structured responsibility, and civil liability can reinforce it by treating the absence of sound governance as a failure of care. Without that operational lens, responsibility will continue to be assigned at the margins, while the central sources of risk remain largely untouched.

B. THE SRE PARADIGM AND THE CULTURE OF RELIABILITY

Site Reliability Engineering emerged from attempts to make complex technical systems more manageable, but its lasting influence has less to do with the specific tools it introduced and more with the way it reshaped responsibility. When a team starts tracking indicators that reflect real user experience, defines service thresholds in clear terms and accepts that a certain margin of failure is unavoidable, something changes. A shared vocabulary begins to take shape across development, operations and management.

And once that vocabulary settles in, the tone of the conversation shifts.

Failure stops being treated as an exception and becomes something closer to a working condition. Discussions become more practical, sometimes almost routine, focused on what can be changed without undermining the parts of the system that actually matter. Reliability, at that point, is no longer an optimistic assertion. It is something that has to be demonstrated, and usually more than once.

The error budget illustrates this shift quite clearly. It is often described as permission for risk-taking, although in practice it tends to operate more as a constraint. As consumption rises, deployments slow down. Once the threshold is crossed, they stop altogether unless there is a convincing reason to continue.

In theory, at least.

Everything depends, in the end, on the honesty of the metrics. If indicators are flattering rather than descriptive, risk accumulates quietly and tends to surface when it is least expected. Scholars in Latin America have been pointing to this problem for some time. A system that chases numbers for their own sake is no safer than one that keeps no metrics at all. It simply becomes harder to audit and more difficult to hold accountable once something goes wrong (Melo, 2022; de Teffé and Medon, 2020).

Post-incident reviews conducted without assigning blame are often framed as cultural practices, but their significance is also institutional. When carried out with some care, they produce a record of how the system behaved, which assumptions failed and why certain remedial actions were taken. A thorough review can serve as evidence of diligence, even when the incident itself caused disruption. It shifts responsibility away from individual reactions and places it in documented practices of design, monitoring and correction. This aligns with regional legal doctrines that link responsibility to duties of prevention, transparency and continuous response, rather than to the simple absence of reported failures (Doneda, Mendes and de Souza, 2020; Bioni and Dias, 2020).

For SRE to acquire real normative weight, a minimum evidentiary baseline has to be available at any given time. A service should be able to present the indicators that guide its decisions, explain how those indicators relate to user outcomes, define acceptable thresholds and identify the points at which human intervention becomes necessary. It should also be able to show how much of its error budget has been used and what recent incident reviews have actually concluded.

If that material is missing, the rest becomes hard to take seriously.

These elements help distinguish organisational structure from empty rhetoric. Their absence, on the other hand, usually signals that SRE has not moved much beyond vocabulary. This tends to be particularly visible in Latin American contexts, where data governance and user protection regimes increasingly rely on objective organisational duties. The transparency expected in SRE mirrors legal expectations of traceability and helps give substance to a standard of reasonable care that is demonstrated through procedure rather than declared through intent (Bioni and Dias, 2020).

A recurring difficulty, however, is that some organisations adopt SRE language without changing the internal conditions that make honest prioritisation possible. Dashboards

multiply, yet the necessary conversations about what actually matters never quite happen. Reliability becomes something that is performed rather than practiced.

And that difference is not trivial.

It only becomes real when metrics start to have consequences. If the error budget is exhausted, new features do not ship. If recovery mechanisms reveal fragility, redesigns are postponed even when they are strategically attractive. These decisions are costly, and not always popular, but they are also the moments in which reliability becomes tangible. They are, in a way, the points at which the law can begin to recognise a form of technical diligence that has practical value rather than decorative appeal. Work on observability and operational risk across the region delivers a fairly direct message. Transparency without the capacity to act achieves little, and the capacity to act without any effort to learn tends to reproduce the same failures.

One of the more interesting shifts introduced by SRE is the way it reframes negligence. The question moves away from identifying who pressed the wrong key and towards examining whether the system was designed to fail safely. Evidence is needed to show that failure was anticipated, kept within manageable bounds and made recoverable.

That is a different standard altogether.

Under this view, negligence lies not in the existence of error but in the absence of structures that detect problems early, interpret them with evidence and correct them in a stable way. This perspective brings technical practice closer to Latin American approaches that understand responsibility as a continuous task of governance, where prevention and remediation belong to the same cycle and diligence is assessed through documents, traces and operational processes rather than through assurances of quality (de Teffé and Medon, 2020).

C. LAW OF TECHNOLOGICAL RISK AND RESPONSIBILITY IN AUTOMATED ENVIRONMENTS

Legal systems have struggled for years to keep pace with technologies that learn, adapt and make decisions without waiting for anyone's approval. Once algorithms begin producing outcomes that even their own developers cannot fully anticipate, the older

framework built around authorship, causation and blame starts to lose part of its explanatory force. Law continues to search for a clear actor behind each decision, yet automated systems distribute agency across models, code, data pipelines and the infrastructure that supports them.

The problem is not particularly subtle.

What is needed is less the creation of a new doctrinal category and more a practical way of treating technological risk as something to be managed rather than eliminated. Canada's PIPEDA already asks organisations to shield individuals from the effects of automated decisions, and the GDPR in Europe expects some form of accountability for algorithmic processes. These instruments differ in method, but they are clearly responding to similar pressures (de Teffé and Medon, 2020).

Short of that, supervision becomes largely interpretative.

In these environments, risk becomes part of routine operation instead of a rare anomaly. The relevant question shifts, and not always comfortably for legal analysis. It is no longer whether systems will fail, because they will.

The harder question comes later.

How responsibility is shared when they do, who recognised the problem early and who ultimately bears the cost when the operational chain breaks. Engineering disciplines recognised this pattern earlier and developed mechanisms to cope with error without systemic collapse.

And law is still catching up.

Legal analysis is adapting, although more slowly and sometimes unevenly. Diligence can no longer be assessed by asking how many failures were avoided. It has to be judged by whether institutions took reasonable steps to anticipate foreseeable harm and by how they reacted when behaviour deviated from what was expected. This reflects a move toward an ex ante understanding of responsibility, where the emphasis falls on managing risk in a timely and reasonable way rather than on the simple absence of incidents (de Teffé and Medon, 2020).

Another aspect of this shift is the gradual move away from individual fault and toward institutional responsibility. Automated systems are collective by design. Developers,

operators, data providers, cloud platforms and the models themselves each contribute, sometimes in ways that are not easy to disentangle, to the final outcome. Searching for a single culprit rarely produces clarity and, in many cases, adds very little from a legal perspective.

In practice, those lines are rarely clean.

Attribution tends to work better when it turns to governance. Policies that define control, records of relevant decisions, traceable data flows and mechanisms for correction often provide a more reliable picture of technical care than any attempt to identify the individual who happened to interact with the system last. This perspective mirrors the culture of reliability in SRE, where responsibility is recognised in organisational routines rather than in the absence of isolated errors. For law, the lesson is not to replicate engineering tools but to accept that error forms part of the ordinary behaviour of complex systems and has to be managed accordingly (Doneda, Mendes and de Souza, 2020).

In production environments, the absence of logs is rarely neutral. It usually means that something failed before anyone was able to observe it.

Latin American scholars have pushed this line of reasoning further. They have shown how automation can magnify existing inequalities, leaving vulnerable groups exposed to systems they cannot meaningfully challenge. Any credible account of technological risk has to take these distributive effects seriously. The point extends beyond the region. Canadian data protection law acknowledges it through safeguards on automated decisions, and the GDPR's emphasis on transparency and accountability reflects a similar concern.

In practice, harm tends to concentrate where protections are weakest.

A law of technological risk that deserves the name must therefore balance efficiency with fairness, ensuring that the burdens created by automation do not systematically fall on those with the least capacity to contest them (Melo, 2022; Bioni and Dias, 2020).

D. REDEFINING “FAILURE” AND “NEGLIGENCE” THROUGH THE SRE LENS

Engineers tend to speak about failure with a matter-of-fact clarity that legal analysis has long struggled to match. Within Site Reliability Engineering, failure is not treated as a

dramatic rupture in an otherwise stable system. It is one of the predictable costs of operating large services that react to fluctuating demand, imperfect configurations and sudden changes in user behaviour. A spike in traffic can expose a flawed assumption in a scaling rule. A deployment script that has worked for months can fail at the one moment it matters. None of this usually surprises an SRE team. The relevant question is whether the organisation has designed its processes so these slips are detected early and their impact contained.

This logic translates relatively directly to law. A failure is not automatically a legal problem. It becomes one when the organisation lacks a structured method for identifying the issue, analysing its causes and preventing recurrence. The global outage affecting Facebook, WhatsApp and Instagram on 4 October 2021 illustrates the point quite clearly. A faulty configuration change disconnected Meta's backbone routers, propagating a failure that took its services offline for more than six hours. The firm later published a detailed post-mortem explaining the sequencing of events, the safeguards that failed and the remedial measures taken. From a legal perspective, that record functions as evidence of diligence: an institution acknowledging error, producing a traceable explanation and showing how it incorporated lessons for the future (Meta Engineering, 2021). By contrast, repeated failures without any documented attempt to learn from them point more toward institutional neglect than simple misfortune.

A similar contrast was visible in the massive disruption suffered by Mercado Pago on 23 November 2022. Users across Latin America were unable to complete payments or transfers for several hours. The company confirmed service degradation but provided only a brief public statement and released no technical report that might shed light on root causes, safeguards or follow-up measures. The absence of documentation does not necessarily imply wrongdoing, but it does limit the ability of regulators to understand the event and assess whether the response met a reasonable standard of care (Infobae, 2022). Vodafone España's nationwide outage on 4 February 2020 raised comparable concerns. Although the company acknowledged the incident and restored service gradually, public reporting revealed limited detail about why the system failed or how it planned to prevent similar events (Muñoz, 2020). The evidentiary gap becomes relevant when responsibility depends not on the occurrence of failure but on how the organisation actually responded.

The error budget sharpens this distinction. It sets an explicit tolerance for degradation and requires teams to slow or halt deployments when instability grows. Engineers often treat it

as a limit on risk rather than a licence to take it, even if in practice that line is not always perfectly clear. In legal terms, it translates abstract duties of foresight into specific decisions about when to deploy, when to roll back and when to stabilise. If a team delays an update because the error budget is nearly consumed, it is following a documented assessment of risk. The focus therefore shifts from isolated mistakes to the quality of the organisation's monitoring and decision-making structures, which aligns with regulatory approaches that tend to prioritise prevention over punishment (Calo, 2021).

Seen from this perspective, negligence has little to do with the actions of one inattentive engineer. It reflects the failure of an organisation to build the routines needed to detect problems, track them and resolve them. A service that loses orders because no one checks error logs is not simply suffering from bad fortune. It is operating without adequate governance.

That difference tends to show up quickly in production.

The first mistake may be understandable. The refusal, or inability, to learn from it is not. Scholars have stressed that accountability must be viewed as a structural rather than individual question, and automated systems make that shift difficult to avoid (Smuha, 2020).

Blameless post-incident reviews reinforce this approach. They are not exercises in assigning blame but structured efforts to reconstruct how the system behaved, how the failure surfaced and how recovery unfolded. When documented with care, they perform a function similar to a legal record. They establish a narrative of cause, evidence and remedy that can be examined by regulators. When no such record exists, it becomes easier to infer negligence, even if the underlying technical issue was not extraordinary. The absence of documentation rarely means that nothing happened. More often, it suggests that the institution did not look closely enough, or chose not to.

The idea of controlled failure also challenges earlier assumptions in legal doctrine. Not every deviation signifies fault. Some failures are expected and, in a limited sense, even planned. Systems absorb them without major harm because safeguards exist. A brief interruption during maintenance is not comparable to an untested deployment that disables a banking platform. The distinction turns on preparation and response, not on the simple fact that something failed. Traditional doctrine, which tends to equate malfunction with blame, struggles to hold up in the context of distributed architectures.

Reasonable technical care therefore demands a different understanding. It is not demonstrated by the absence of incidents but by the ability to maintain documentation, review assumptions, strengthen safeguards and incorporate lessons into routine practice. Reliability becomes less a promise and more a way of working. Engineering provides the artefacts that make this visible. Law, in turn, provides the expectation that these artefacts must reflect consistent and meaningful practice.

In this sense, SRE offers more than useful vocabulary. It provides a way of approaching failure that becomes difficult to ignore once systems reach a certain level of complexity. The question is no longer simply whether something went wrong, but whether anyone was in a position to notice it and respond in time.

That is where the legal problem actually sits.

Diligence, in practice, depends on whether the organisation understood what happened, recorded it and acted on it in a way that can be examined later. Without that, the distinction between error and negligence becomes harder to sustain.

Some systems fail. Others are left to fail.

The difference is not technical.

E. REGULATORY IMPLICATIONS AND CHALLENGES FOR LAW

Regulation has often assumed that technical systems become manageable once the right rules are written, as if control could be imposed from the outside. Automated technologies make that assumption increasingly difficult to sustain. Engineers have long understood that complex systems fail in ways that are neither linear nor fully predictable.

The problem is not lack of rules.

The Facebook–WhatsApp–Instagram global outage of 4 October 2021 made this visible on a public scale. A faulty configuration change propagated across Meta’s backbone network and removed the Border Gateway Protocol routes that allowed the company’s systems to be reachable on the internet. For nearly six hours, global communication and commerce were disrupted despite the absence of a malicious attack. The incident showed how quickly an

internal misconfiguration can cascade outward, often beyond the reach of conventional oversight (Frier, 2021).

What failed was not a control, but a sequence.

Legislating against every possible error achieves little. What matters, in practice, is whether deviations are detected early, contained effectively and used to guide learning rather than denial. Aviation accepted this logic long ago by designing layered safeguards that begin from the expectation of error. Modern software systems, which can propagate faults within seconds, make that lesson difficult to ignore (Smuha, 2020).

A regulatory approach grounded in reliability therefore pays less attention to how often an organisation stumbles and more to whether it understands what the stumble meant. The large outage that affected Vodafone España on 4 February 2020, which left thousands of users without service, illustrates the point. The company identified routing failures in its infrastructure and adjusted traffic management practices accordingly, producing a form of operational learning that no static compliance checklist could have generated (Muñoz, 2020).

That kind of learning is uneven.

Law has to move from inspecting outcomes to examining processes. Site Reliability Engineering's emphasis on routine error management shows how mistakes can become sources of insight rather than automatic triggers for blame (Hood, Rothstein and Baldwin, 2001).

Regulators in different parts of the world are beginning to converge on this idea, even if their policy instruments differ. The European Union has recently adopted DORA for financial services and continues to refine the proposed AI Act. In the United Kingdom, the Information Commissioner's Office used the 2018 British Airways data breach to demonstrate that accountability requires evidence of monitoring, documentation and response. Although the initial fine proposed by the regulator was later reduced, the case showed how supervisory bodies treat failures in system configuration, logging and security alerts as indicators of broader governance weaknesses rather than isolated mistakes (Information Commissioner's Office, 2020).

In practice, convergence does not mean consistency.

In the United States, the release of NIST's AI RMF 1.0 reflects an attempt to provide a shared vocabulary for assessing AI risk. Singapore's 2020 governance framework remains

influential in Asia, while Japan and South Korea continue to experiment with their own models of oversight. Across Latin America, financial supervisors are beginning to explore digital resilience models that prioritise process over static inspection. These trends appear to be moving in a similar direction, although unevenly.

The central challenge is cultural as much as legal. Many legal systems still equate error with wrongdoing. Blameless reviews can therefore feel counterintuitive, yet they often provide insights that formal investigations fail to capture.

In production, silence is rarely a good sign.

An online marketplace that identifies an overloaded server as the cause of an outage and updates its load-testing practices has shown diligence in a way that classical doctrine might overlook. By contrast, punitive environments tend to push organisations toward concealment or minimal disclosure. A supervisory model that rewards clear and timely incident reporting is more likely to encourage improvement than evasion (Smuha, 2020).

A mature law of technological risk would create conditions in which organisations can disclose problems without immediately fearing disproportionate penalties. This is not an argument for leniency. It is, more precisely, an argument for learning. A cloud service that publishes a detailed account of an outage offers greater protection to its users than one that simply pays a fine while revealing little. Public scrutiny of incident logs allows a more grounded assessment of whether institutions actually understand their systems and whether they have addressed the underlying causes of failure (Hood, Rothstein and Baldwin, 2001).

For this to work, regulators themselves need to evolve. Oversight cannot rely solely on doctrine. It requires a working familiarity with code, system architectures, operational behaviour and the evidentiary value of logs and metrics. Paper-based inspections are of limited use when systems change multiple times per hour.

Supervision cannot remain external.

Regulators increasingly need to work alongside engineers to translate technical artefacts into supervisory judgment. Europe has already begun training regulators in algorithmic audits and AI impact assessment. Singapore's risk-focused audits push supervisors to understand system behaviour more directly. Latin America, although slower to formalise these practices, benefits from administrative traditions that tend to adapt pragmatically to evolving technologies (Doneda, Mendes and de Souza, 2020).

Ethics underpins this entire discussion. Treating error as a normal feature of automation must not become an excuse for careless design or unsafe practices. Controlled failure only makes sense when users are protected and when tolerance remains within socially acceptable boundaries.

That boundary is not always clearly defined.

Regulation should not become paralysed by an unattainable vision of perfection, but nor should it slide into permissiveness. The task is to strike a balance that maintains trust. In that balance, law can reposition itself not as an obstacle to innovation but as a necessary condition for it. A culture of reliability built around shared responsibility, institutional learning and honest repair sits, in practice, at the intersection between technical logic and the demands of justice.

F. CONCLUSIONS

The relationship between technological reliability and legal responsibility is no longer an abstract discussion. It unfolds daily in systems that fail while real users watch and in institutions that must decide whether those failures reflect ordinary fragility or something else entirely.

Sometimes that distinction is not obvious at all.

Engineers have spent years refining methods to keep services stable under pressure and to extract lessons from the moments when stability breaks. Law is beginning to recognise that its own legitimacy depends on something similar. General principles still matter, but they no longer suffice on their own. What is increasingly required is verifiable evidence of how systems actually behave in practice. Site Reliability Engineering offers one practical route into that evidentiary discipline by insisting that infrastructures must be observable, measurable and open to scrutiny, conditions that also underpin effective cybersecurity practices (NIST, 2023; European Commission, 2018).

Recent regulatory developments point in the same direction, although not always in a perfectly aligned way. The publication of NIST's AI RMF 1.0 signals a willingness to evaluate automated decision-making in more concrete terms. Canada's PIPEDA ties accountability to meaningful explanations. Europe's GDPR continues to push

organisations to confront the opacity of their systems more directly, even if implementation varies in practice. Singapore's 2020 framework adds a model of supervision organised around risk.

None of these instruments promises the elimination of failure.

And perhaps they shouldn't.

Once responsibility is analysed through the lens of reliability, the legal timeline begins to shift. The assumption that harm comes from a single discrete event becomes harder to maintain in environments where systems generate continuous micro-decisions and where harm often emerges gradually. Continuous observation becomes central to supervision. Diligence stops being an impressionistic standard and becomes something that can be demonstrated with evidence, not just asserted. Error budgets, service indicators and post-incident reviews move from engineering artefacts to legal evidence, including in the analysis of security incidents. They become records of what an organisation understood, when it understood it and how it responded.

That shift is not minor.

Latin American scholarship adds an ethical dimension that is sometimes underestimated elsewhere. In societies marked by inequality, the consequences of technical failure do not fall evenly. Reliability becomes not only a question of design but also of fairness. This intuition travels well beyond the region. Whether in European guidance, NIST's recommendations or OECD principles, a similar concern keeps appearing, even if framed differently. Public trust grows through transparent practices rather than through increasingly dense rulebooks (NIST, 2023; OECD, 2019; European Commission, 2020).

A law of technological risk should resist the temptation to force systems into predictable shapes. It has to work with uncertainty. Not eliminate it.

Engineering shows that resilience is built over time through design, monitoring and learning, often in small iterations rather than through grand redesigns. Law can reinforce this by requiring documentation, traceability and audits that reveal how institutions respond to error in practice. The same applies to cybersecurity, where the absence of traceable evidence often obscures both cause and responsibility, sometimes long after the incident itself.

The aim is not heavier control. It is better memory.

An organisation that keeps and analyses its incident history is not just avoiding negligence. It is, in a more practical sense, building a form of trust that legal systems can recognise and, eventually, protect.

The most significant change ahead is likely to be cultural. Accepting that systems fail does not justify irresponsibility. It reflects a more realistic understanding of limits, both human and technical. Responsibility shifts from preventing every failure to responding to failures with clarity, discipline and a certain degree of honesty.

A documented failure tells a story.

A concealed one does too.

If contemporary risk theory teaches anything, it is that SRE offers a useful guide for the legal transition now underway. It encourages regulators to focus on learning rather than control for its own sake. It suggests that the aim is not to eliminate harm but to convert experience into knowledge and to maintain supervision that adapts as systems evolve.

Law does not need to outrun technology.

But it does need to understand it.

Only then will legal institutions be prepared to govern the systems upon which society increasingly depends (Hood, Rothstein and Baldwin, 2001).

REFERENCES

Beyer, B., Jones, C., Petoff, J., & Murphy, N. (2016). *Site reliability engineering: How Google runs production systems*. O'Reilly Media.

Bioni, B., & Dias, D. (2020). Responsabilidade civil na proteção de dados pessoais. *Civilistica.com*, 9(3), 1–23.

Calo, R. (2021). Liability for robots and other agents. *Annual Review of Law and Social Science*, 17, 105–123. <https://doi.org/10.1146/annurev-lawsocsci-101620-013809>

Cámara Nacional de Apelaciones en lo Civil, Sala D. (2018). *Asociación de Bancos de la República Argentina c/ IBM Argentina SA y otro s/ daños y perjuicios* (AR/JUR/46837/2018). La Ley Online.

de Teffé, C. S., & Bodin de Moraes, M. C. (2017). Redes sociais virtuais: Privacidade e responsabilidade civil. *Pensar*, 22(1), 108–146. <https://doi.org/10.5020/2317-2150.2017.6084>

de Teffé, C. S., & Medon, F. (2020). Responsabilidade civil e regulação de novas tecnologias: Reflexões sobre a tomada de decisões com sistemas de inteligência artificial. *Revista de Estudos Institucionais*, 6(1), 151–182. <https://doi.org/10.21783/rei.v6i1.437>

Doneda, D., Mendes, L. S., & de Souza, C. A. P. (2020). Considerações iniciais sobre inteligência artificial, ética e autonomia pessoal. *Pensar*, 23(4), 1–20. <https://doi.org/10.5020/2317-2150.2020.9808>

European Commission. (2018). *Regulation (EU) 2016/679 (GDPR)*. <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32016R0679>

European Commission. (2020). *Proposal for a regulation on digital operational resilience for the financial sector (DORA)*. <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52020PC0595>

European Commission. (2021). *Proposal for a regulation laying down harmonised rules on artificial intelligence (Artificial Intelligence Act)*. <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52021PC0206>

European Commission. (2022). *Regulation (EU) 2022/2554 on digital operational resilience for the financial sector (DORA)*. <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32022R2554>

Forsgren, N., Smith, D., Humble, J., & Frazelle, J. (2021). *Accelerate: State of DevOps 2021*. Google Cloud & DORA.

GitHub Engineering. (2018, October 30). *October 21 post-incident analysis*. GitHub Blog. <https://github.blog/>

Government of Canada. (2000). *Personal Information Protection and Electronic Documents Act (PIPEDA)*. <https://laws-lois.justice.gc.ca/>

Hood, C., Rothstein, H., & Baldwin, R. (2001). *The government of risk: Understanding risk regulation regimes*. Oxford University Press.

- Infobae. (2022, November 23). *Se cayeron Mercado Libre y Mercado Pago: los usuarios reportaron problemas para operar con la plataforma*.
<https://www.infobae.com/economia/2022/11/23/se-cayeron-mercado-libre-y-mercado-pago-los-usuarios-reportaron-problemas-para-operar-con-la-plataforma/>
- Infocomm Media Development Authority. (2020). *Model artificial intelligence governance framework* (2nd ed.). Singapore Government.
- Kubo, M. (2022). Ethical and legal challenges in AI deployment: A Japanese perspective. *Journal of AI Ethics*, 2(3), 215–230. <https://doi.org/10.1007/s43681-021-00104-8>
- Melo, B. L. D. A. A. (2022). Sistemas de inteligência artificial e responsabilidade civil: Dificuldades dos modelos tradicionais. *Revista de Direito da Faculdade de Jataí*, 6(1), 1–26.
- National Institute of Standards and Technology. (2023). *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*. <https://doi.org/10.6028/NIST.AI.100-1>
- Organisation for Economic Co-operation and Development. (2019). *Recommendation of the Council on Artificial Intelligence*. OECD Publishing.
<https://legalinstruments.oecd.org/en/instruments/OECD-LEGAL-0449>
- Palazzi, P. (2012). Responsabilidad de los intermediarios en Internet. *Revista de Derecho Privado y Comunitario*, 2012(1), 45–68.
- Siddiqui, F. (2021, October 5). *Facebook, Instagram and WhatsApp suffer global outage*. The Washington Post.
<https://www.washingtonpost.com/technology/2021/10/04/facebook-down-instagram-whatsapp/>
- Vodafone outage — Spain. (2020, February 19). In R. Muñoz, *Una avería deja sin servicio a miles de clientes de Vodafone*. *El País*.
https://elpais.com/economia/2020/02/19/actualidad/1582102382_369209.html